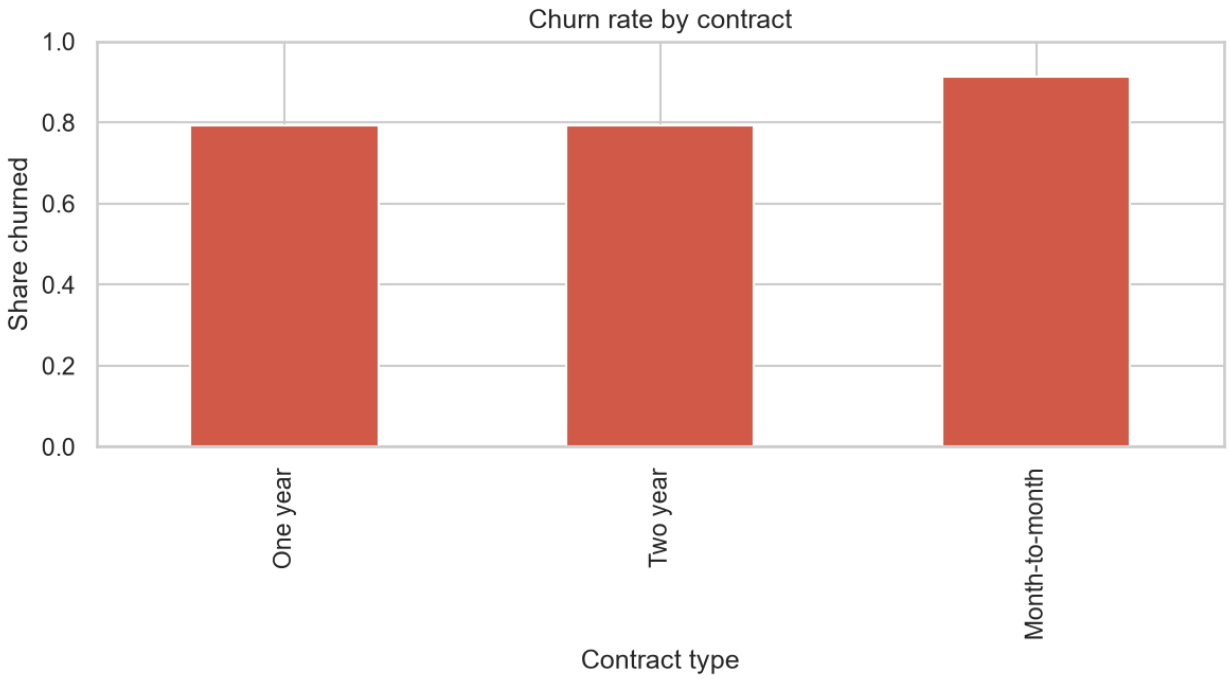
Customer Churn Risk and Retention Analysis



Prepared: 2026-10-20 | Rows: 1,200 | Churn rate: 86.3% (1,036/1,200)

Objective: rank customers for a constrained retention contact list and explain the resulting risk signals.

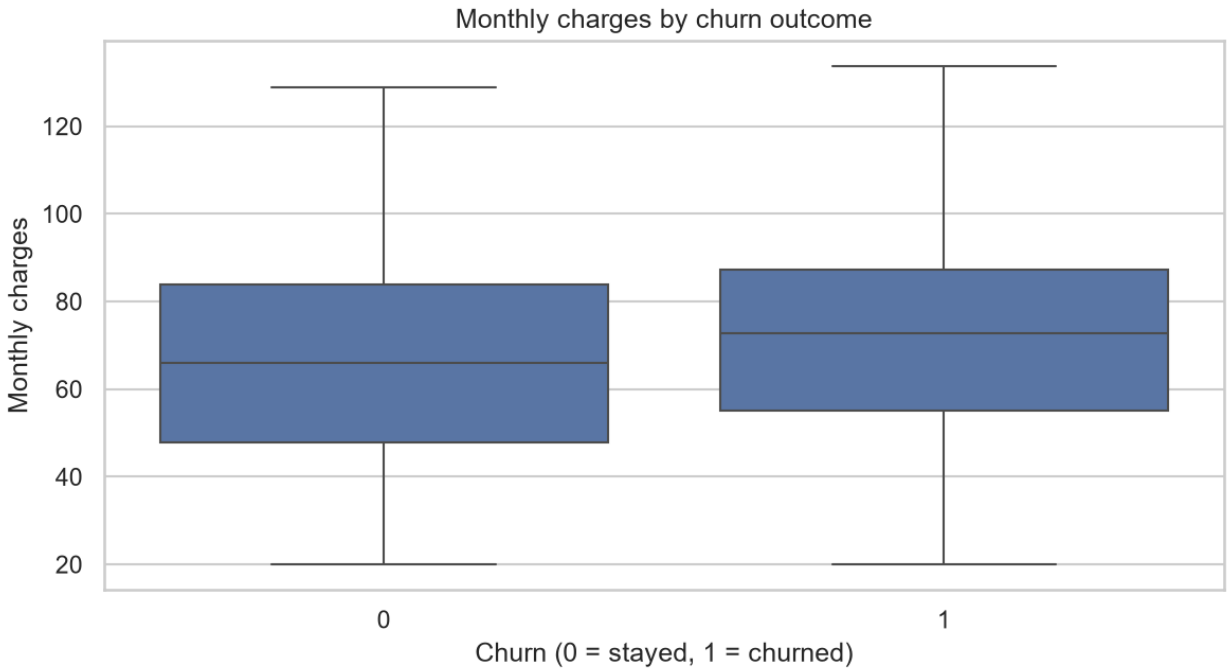
Data is a single labeled snapshot. Associations are predictive, not causal, no intervention outcomes are supplied.

Selected model: RandomForest

Evaluation uses a stratified 75/25 split (seed 42) the test set was held out during tuning.

Business assumptions: contact cost 8, *valueofaretainedcustomer*180, assumed save rate 25%, capacity 20%.

1. Data Audit and EDA



Missing values: monthly_charges=30, total_charges=24, satisfaction_score=48, monthly_usage_gb=36.

Exact duplicate rows: 0, repeated customer IDs: 0.

Quality checks (out-of-range / invalid counts): age_out_of_range=0, tenure_out_of_range=0,

satisfaction_out_of_range=0, negative_charges=0, negative_complaints_or_late_payments=0,

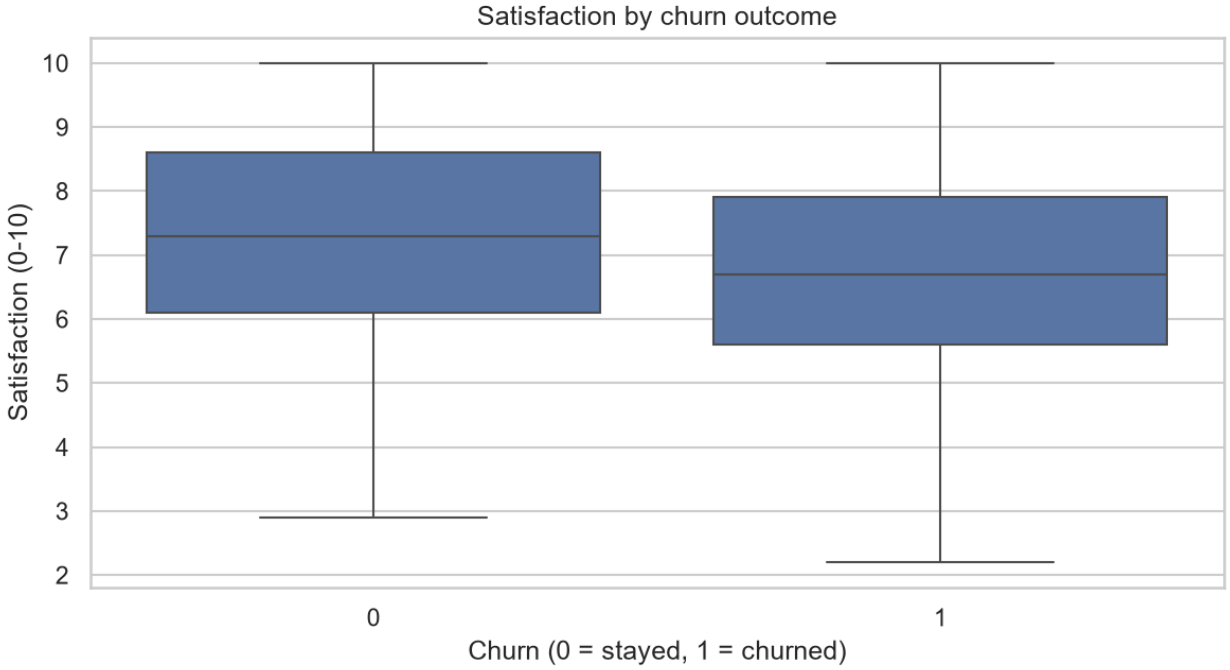
discount_out_of_range=0, unexpected_binary_labels=0, target_invalid=0

Cleaning: no rows removed; raw missing values retained in cleaned CSV. Numeric and categorical imputation is fitted within each training fold.

Outliers were reviewed with robust plots and ranges, not trimmed. High usage/charge values can be legitimate, confirm with source-system rules before correction.

customer_id is excluded from all predictive features. Churn: 86.3% positive, so accuracy and majority-class baselines are misleading.

2. EDA: Customer Signals



The figures summarize contract, tenure, charges, satisfaction, complaints, and late payments. Payment method, usage/charge, and contract/service plots are included in the notebook figure set.

Observed churn was 91.2% for month-to-month contracts and about 79% for one- and two-year contracts. Median satisfaction was 6.7 among churners and 7.3 among non-churners, median monthly charge was 72.78 *vs* 66.09.

Churn rose to 93% for customers with 2+ late payments, while complaint counts of 3+ were high-risk but sparse. Treat these small-slice rates cautiously.

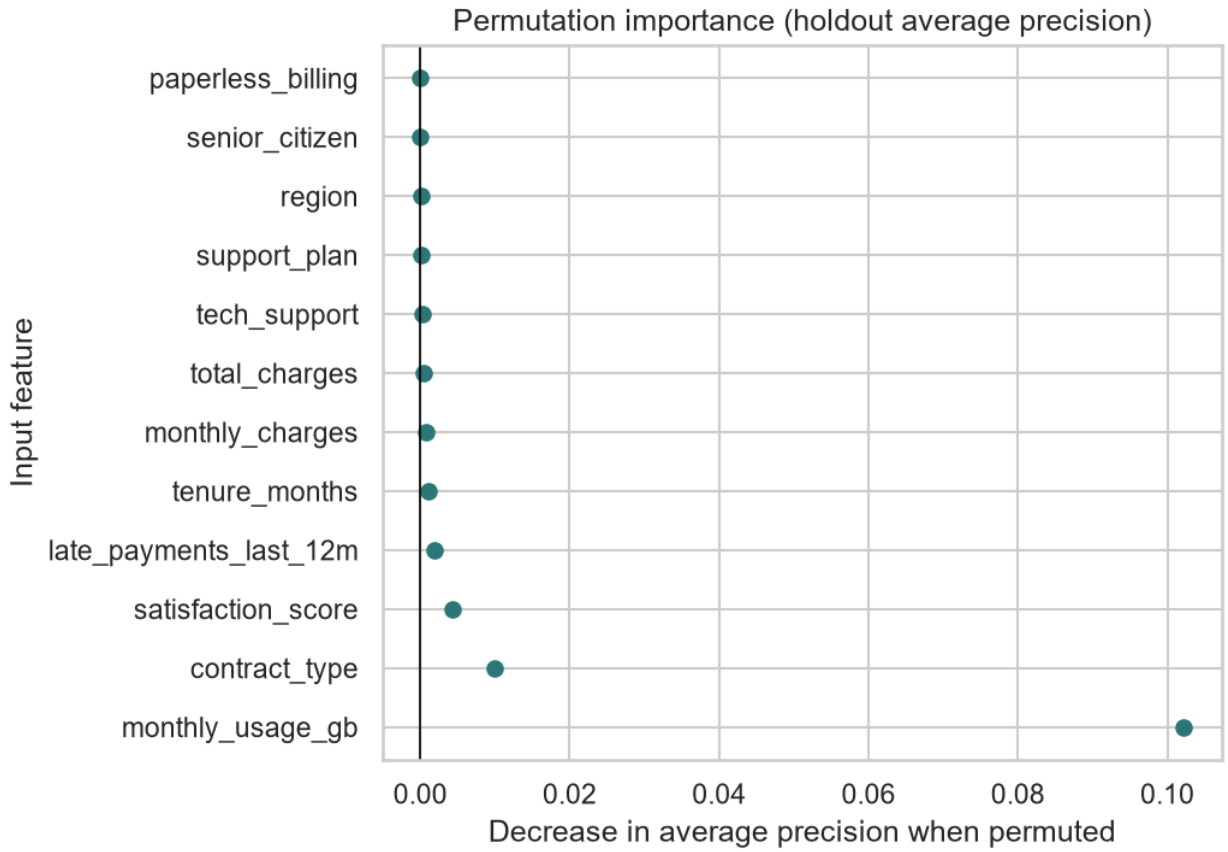
Interpret subgroup differences as descriptive associations, small groups and the high churn prevalence can make raw rates unstable.

Engineered fields: average charge per tenure month, payment-risk score, complaint intensity, service friction, low-satisfaction flag, high-cost/low-usage ratio, tenure-charge interaction, and month-to-month indicator.

All engineered fields use customer attributes available before the prediction cycle, the target and identifier are never inputs.

Rationale: encode accumulated burden, payment friction, service dissatisfaction, and cost/engagement mismatch without using future outcomes.

3. Modeling and Selection



RandomForest: baseline CV AP 0.972 +/- 0.003; tuned CV AP 0.977. Params: {'model__max_depth': 8,

'model__max_features': 'sqrt', 'model__min_samples_leaf': 2, 'model__n_estimators': 250}

HistGradientBoosting: baseline CV AP 0.972 +/- 0.006; tuned CV AP 0.975. Params: {'model__l2_regularization': 0.0,

'model__learning_rate': 0.05, 'model__max_leaf_nodes': 15}

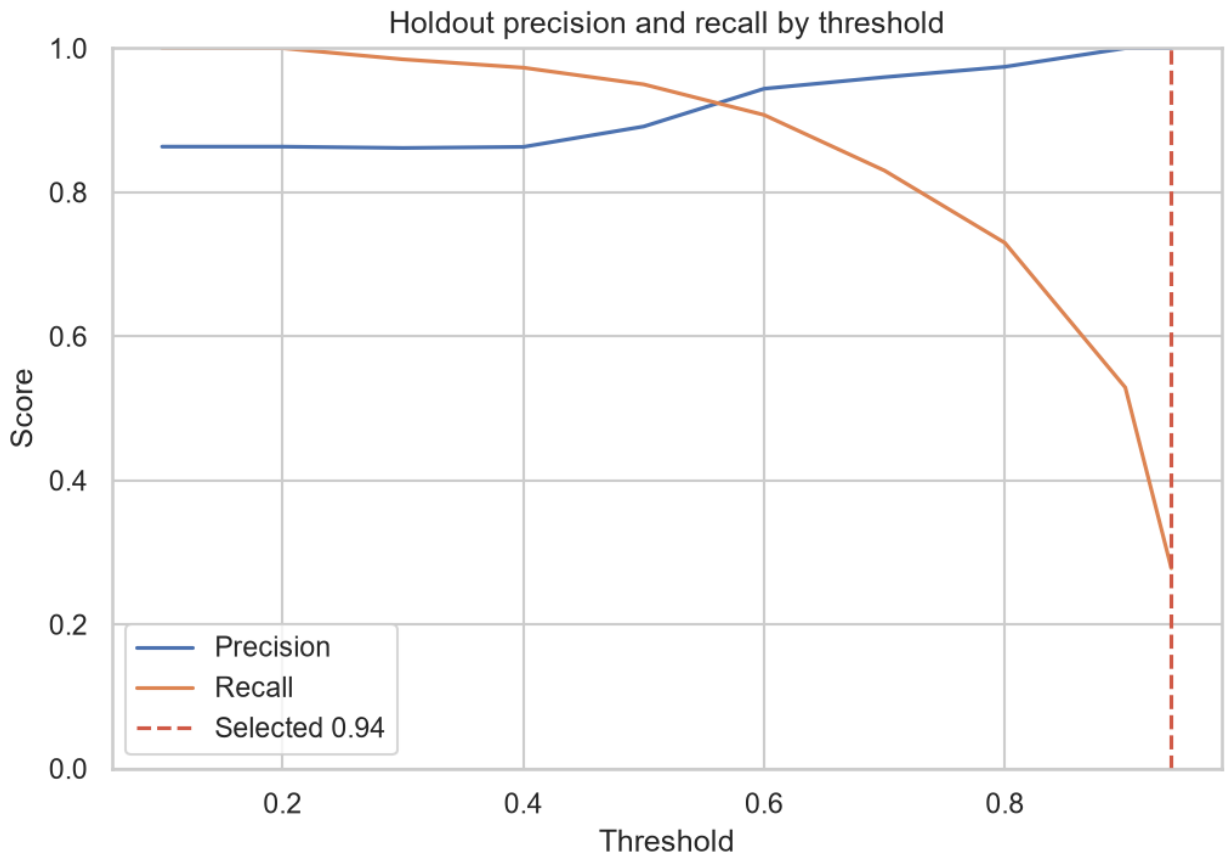
LogisticRegression: baseline CV AP 0.972 +/- 0.004; tuned CV AP 0.974. Params: {'model__C': 0.1}

Search: 4-fold stratified CV; small explicit grids; average precision optimizes positive-class ranking under imbalance.

Operating threshold 0.938 was selected from training-only out-of-fold predictions by maximizing assumed net value under the 20% contact cap.

Treatment: class weights for Logistic Regression and Random Forest, HistGradientBoosting used native unweighted fitting. No resampling, thus no synthetic observations cross folds.

4. Holdout Evaluation and Robustness



Holdout: precision 1.000; recall 0.278; F1 0.435; ROC-AUC 0.890; PR-AUC 0.982; Brier 0.092.

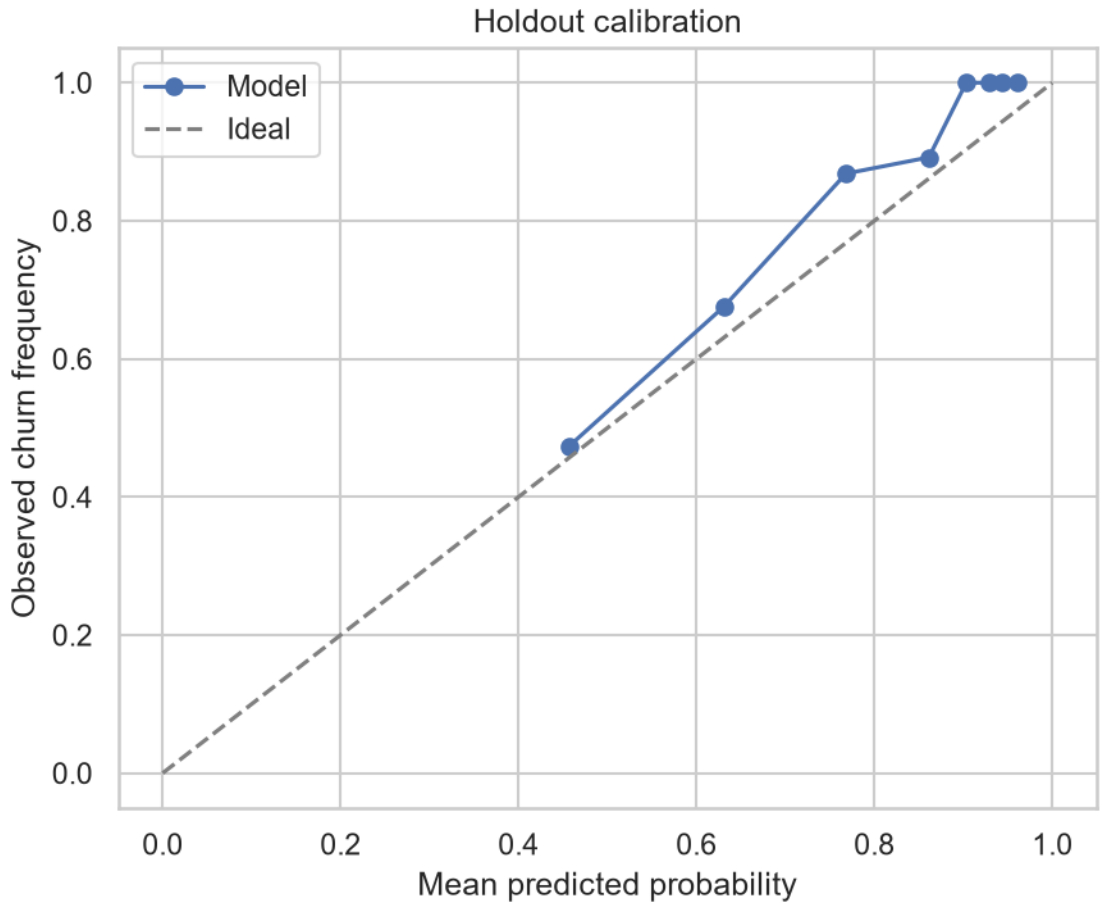
Confusion matrix [[TN, FP], [FN, TP]]: [[41, 0], [187, 72]].

Accuracy alone can look strong when the majority class dominates and does not measure ranking or the value of a limited contact list.

Four-fold training AP variation: mean 0.977, SD 0.001, range 0.977-0.979.

Calibration curve and Brier score assess probability reliability, imperfect calibration is possible, especially in small slices. Threshold and probability should be revalidated on a later time period.

5. Retention Simulation and Explainability



Top 20%: 60 test customers contacted; 60 of 259 observed churners captured (23.2%); random expectation 51.8; lift 1.16x.

Illustrative expected net value: 2, 220; *randomexpected*1,851. These calculations assume a 25% save rate and $180 retained value; actual treatment effect is unknown.

Permutation importance measures how much holdout average precision falls after shuffling each input. It describes predictive reliance, not causal impact.

Three customer-level explanations are provided as one-feature-at-a-time replacement effects relative to training-set typical values. These are local model sensitivities, not additive SHAP values or causal counterfactuals.

Use the ranked list for a measured pilot with randomized treatment/control assignment before claiming incremental retention value.

6. Recommendations and Limitations

Highest permutation signals in this run: monthly_usage_gb, contract_type, satisfaction_score,

late_payments_last_12m, tenure_months. Confirm the direction and stability by subgroup and future cohorts before operational changes.

1. Test proactive contract renewal offers for high-risk month-to-month customers, measure incremental retention in a randomized holdout.

2. Route customers with repeated complaints or weak satisfaction to a service recovery review, with issue-resolution tracking.

3. Offer payment-method or billing support to customers with late-payment signals avoid punitive treatment and assess affordability.

4. Review high-charge / low-usage accounts for plan fit and transparent downgrade options.

5. Monitor risk calibration, contact capacity, subgroup performance, and realized lift each cycle, retrain only after data quality and drift checks.

Limitations: small sample, very high churn prevalence, one snapshot, possible selection artifacts, no causal treatment data, and hypothetical financial values. Validate with temporally separated data before production use.